

# ARUN KUMAR VELUSAMY

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## WORK EXPERIENCE

### Purpleslate Pvt. Ltd. (Cybersecurity Intern)

January 2023-June 2023

- Utilized **Wireshark** to perform deep packet inspection, identifying and filtering suspicious traffic patterns during weekly network audits.
- Documented over **15+ vulnerability findings** per month, contributing to the development of a more robust incident response framework.

### Brilliersys (Database Intern)

July 2022-August 2022

- Optimized SQL queries to handle **50,000+ concurrent transactions**, ensuring database stability during peak usage periods.
- Evaluated system hardware to recommend **RAID 10 configurations**, balancing data redundancy with high-speed performance for enterprise-level data storage.

## EDUCATION

### Master of Science in Cybersecurity

August 2024-May 2026

University of Delaware; GPA: 3.39/4.0

Coursework: Pen testing and Reverse Engineering, Digital Forensics, System Hardening and Protection, Web Application Security, Cloud Computing and Security, Secure Software Design.

### Bachelor of Technology in Information Technology

June 2020-May 2024

Coimbatore Institute of Technology; GPA: 3.1/4.0

Coursework: Cryptography and Network Security / Cyber Security, Computer Networks, Operating Systems, Blockchain & IoT.

## CERTIFICATIONS

- ISC2 Certified in Cybersecurity (CC) 2025
- Cisco Networking Academy® Introduction to Cybersecurity course 2023
- AWS CLOUD PRACTITIONER ESSENTIALS 2023
- secure your cloud data 2022

## SKILLS

**Programming:** Python, C++, C, Java

**Technical tools:** Wireshark, Nmap, Burp Suite, Metasploit, Nessus.

**AI & data science:** Machine learning (logistic Regression, Random Forest, KNN), Predictive Modelling, Pattern Recognition, Large Dataset Handling.

**Cybersecurity:** Network Traffic Modelling, Ethical Hacking, Malware Analysis (Taint/Control Flow), Firewalls, Cryptography.

**Infrastructure:** Windows Server Administration, TCP/IP Networking, AWS, AZURE, RAID systems.

## PROJECTS

### SECURING TRANSACTIONS

- Developed a credit card fraud detection system that processed 280,000+ transactions using Logistic Regression, Random Forest, and KNN algorithms, achieving a 92% precision rate by extracting complex feature patterns and optimizing performance through class weight balancing in imbalanced datasets.

### DYNAMIC TRAFFIC MANAGEMENT AND EMERGENCY OVERRIDE SYSTEM

- Designed an AI-driven infrastructure solution using computer vision to automate signal timings based on real-time density, reducing emergency response times by 20% while validating the technical architecture through a publication in the Mukht Shabd Journal (Impact Factor: 4.6).

### MALWARE DETECTION PRO

- Engineered an automated detection model for Android APK files using control flow and taint analysis to identify malicious patterns, attaining a 94% accuracy rate across a dataset of 1,500+ samples by integrating model checking techniques to refine detection precision.

## INVOLVEMENT

### Sergeant at Arms

Rotaract Club of Coimbatore Smart City

May 2023 - May 2024

- Managed meeting logistics, parliamentary procedures, and professional decorum as a primary co-organizer for Urban Aura 2023, while coordinating with executive members to execute community service initiatives and host over 40 local entrepreneurs.